

K. Satya Sampath Kumar

Software Engineer — MERN Stack & AI-Driven Automation

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Professional Summary

Computer Science Engineering student with hands-on experience building full-stack web applications and automation solutions using React, Node.js, Express.js, and MongoDB. Experienced in developing AI-powered data pipelines, REST API integrations, and automated email notifications, including Retrieval-Augmented Generation (RAG) systems. Passionate about solving real-world business problems through software engineering and continuously strengthening problem-solving skills through Data Structures and Algorithms.

Technical Skills

Languages: Python, JavaScript

Frontend: React, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js

Database: MongoDB, MongoDB Atlas

GenAI: RAG Pipelines, Vector Embeddings, ChromaDB, Ollama

Automation & Integration: REST API Integrations, Email Automation, CI/CD Deployment Automation

Tools: Git, GitHub, VS Code

Projects

Interview AI Platform

[GitHub](#) — [Live Demo](#)

- Developed a full-stack AI interview preparation platform using React, Node.js, Express.js, MongoDB Atlas, and Google Gemini AI.
- Built an automated ATS-friendly resume analysis pipeline that generates personalized questions, interview reports, and skill-gap analysis from uploaded resumes.
- Implemented secure JWT authentication and deployed the application using Vercel, Render, and MongoDB Atlas.

Real-Time Collaborative Editor

[GitHub](#) — [Live Demo](#)

- Built a collaborative code editor supporting simultaneous multi-user editing with conflict-free synchronization using Yjs and Socket.IO.
- Integrated real-time communication with WebSockets and designed a responsive React interface.
- Deployed the application using Render and Vercel.

Education

Bachelor of Technology in Computer Science Engineering

2023 – 2027

Gandhi Institute of Technology and Management (GITAM), Visakhapatnam

CGPA: 8.2

Internship Experience

AI Virtual Internship – PolicyNav | Infosys Springboard

Aug 2025 – Oct 2025

[GitHub](#)

- Built an automated Retrieval-Augmented Generation (RAG) pipeline in Python and Streamlit to simplify navigation of public policy documents.
- Automated document processing using pdfplumber for PDF text extraction and line-based chunking of large documents.
- Generated vector embeddings using Ollama (`nomic-embed-text`) and stored them in a persistent ChromaDB vector database for automated semantic retrieval.
- Built an automated retrieval and grounded-prompting workflow that surfaces relevant chunks using vector similarity and provides source references to reduce hallucinations.

Full Stack Web Developer Intern | SLS Structo-Mech Consultants

Jul 2026 – Present

[slnexus.com](#) — [GitHub](#)

- Developed and maintained the company's official website using the MERN stack to showcase engineering services and completed projects.
- Built an admin dashboard with status tracking (New, To Do, Replied) and delete functionality to manage client inquiries submitted through the website's contact form.
- Integrated Brevo Email API to automate dual email notifications – alerting the site owner with inquiry details (contact number, email, and request type) and sending clients an instant acknowledgment committing to a response within 24 business hours.
- Configured Git-based CI/CD with Vercel and Render for automatic build and deployment on every commit, and managed DNS and hosting setup using Cloudflare.

Achievements & Activities

- Participated in the **Gitathon Hackathon**, collaborating with a four-member team to recreate the **GitHub user interface**.
- Active volunteer in the **National Service Scheme (NSS)**, contributing to community development initiatives.
- Regularly solve Data Structures and Algorithms problems on LeetCode to strengthen coding and problem-solving skills.